



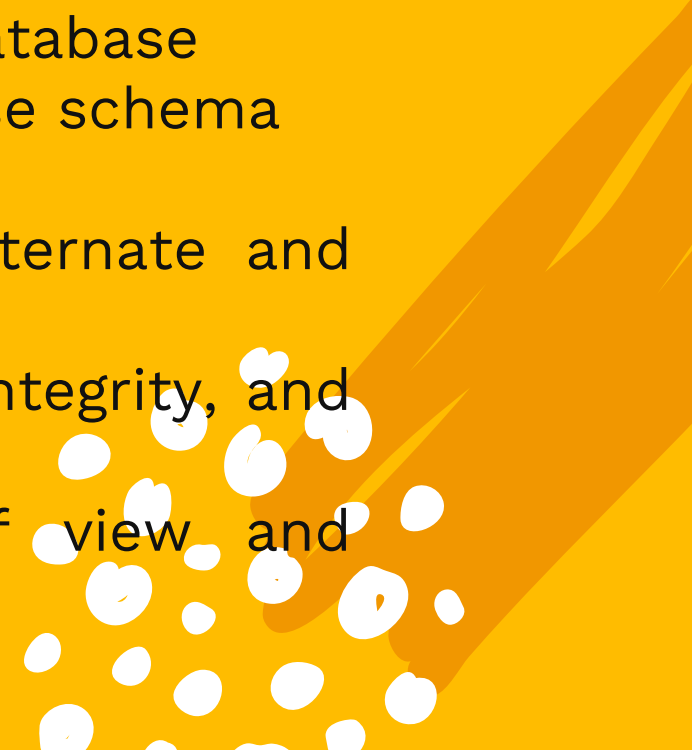
# The Relational Model

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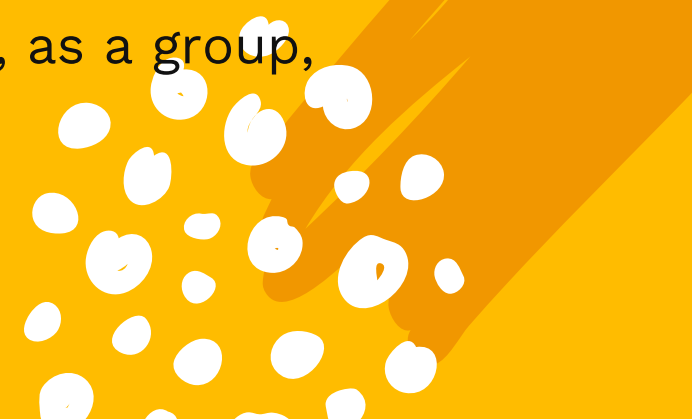
# Objectives

At the end of this lesson, you should be able to:

- X Define relation, attribute, domain, tuple, degree, cardinality, modality and relational database.
  - X Understand the mathematical relation of database
  - X Describe the relation and relational database schema
  - X Explain the properties of relation
  - X Identify the super, candidate, primary, alternate and foreign key
  - X Describe the nulls, entity and referential integrity, and general constrain.
  - X Explain base relation, view, purpose of view and updating views.
- 



# Introduction

- X The relational model was first proposed by E. F. Codd in his seminal paper 'A relational model of data for large shared data banks' (Codd, 1970).
  - X Data appears to be stored in what we have been referring to as **simple, linear files**.
  - X Relational databases are based on **mathematics**.
  - X A relational database is a **collection of relations** that, as a group, contain the data that describes a particular business environment.
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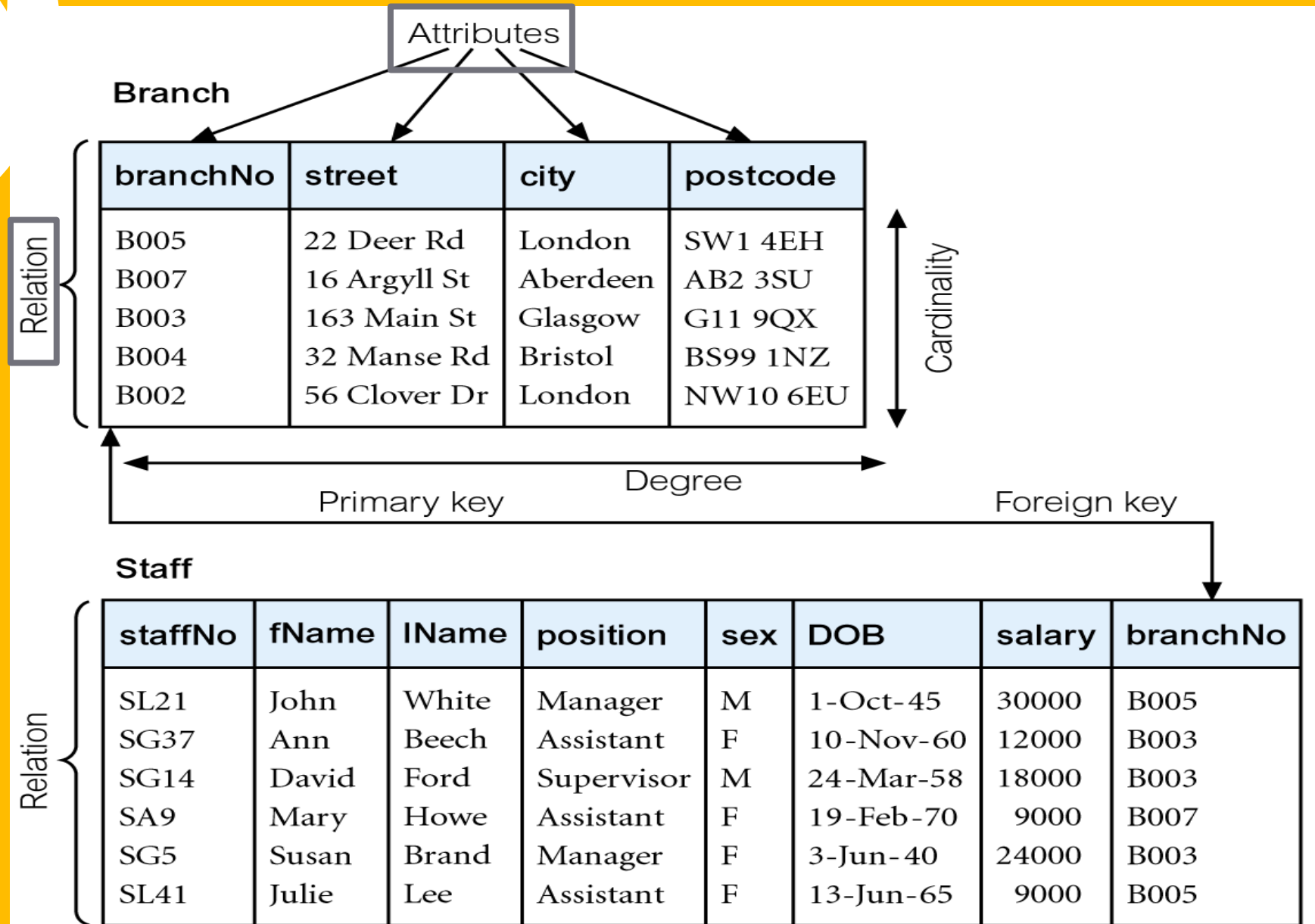


# Terminology



- x The relational model is based on the mathematical concept of a relation, which is physically represented as a table.
  - x Relation : table with columns and rows.
  - x Attribute : Named column of a relation
  - x Domain : The set of allowable values for one or more attributes.
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**Attribute : Named column of a relation**



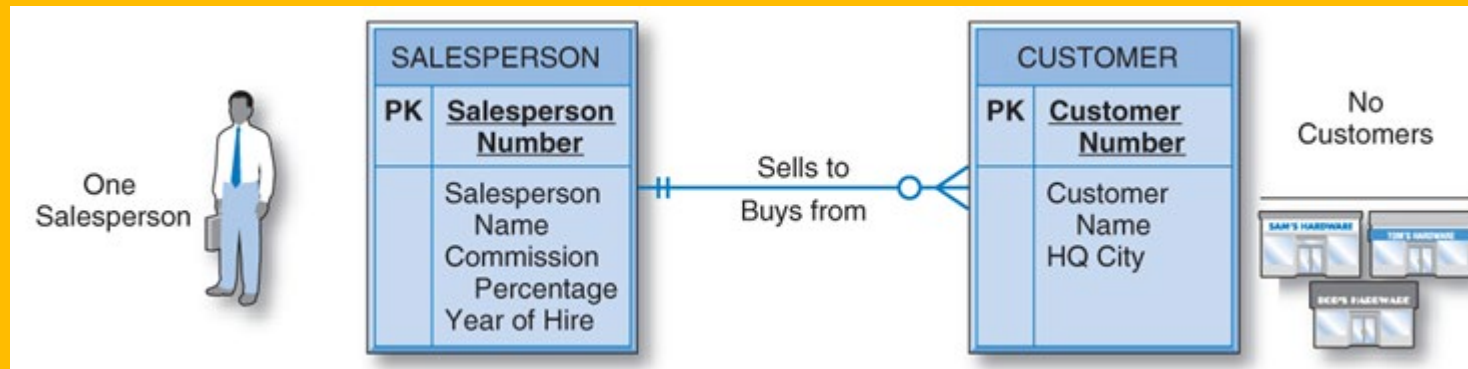
**Relation : table with columns and rows.**

# Example of Attributes Domain

| Attribute | Domain Name   | Meaning                                | Domain Definition                              |
|-----------|---------------|----------------------------------------|------------------------------------------------|
| branchNo  | BranchNumbers | The set of all possible branch numbers | character: size 4, range B001–B999             |
| street    | StreetNames   | The set of all street names in Britain | character: size 25                             |
| city      | CityNames     | The set of all city names in Britain   | character: size 15                             |
| postcode  | Postcodes     | The set of all postcodes in Britain    | character: size 8                              |
| sex       | Sex           | The sex of a person                    | character: size 1, value M or F                |
| DOB       | DatesOfBirth  | Possible values of staff birth dates   | date, range from 1-Jan-20,<br>format dd-mmm-yy |
| salary    | Salaries      | Possible values of staff salaries      | monetary: 7 digits, range<br>6000.00–40000.00  |

# Terminology

- X Tuple is a row of a relation.
- X Degree of a relation is the number of attributes it contains.
- X Cardinality - the maximum number of entities that can be involved in a particular relationship. (Outer)
- X Modality - The minimum number of entity occurrences that can be involved in a relationship. (Inner)

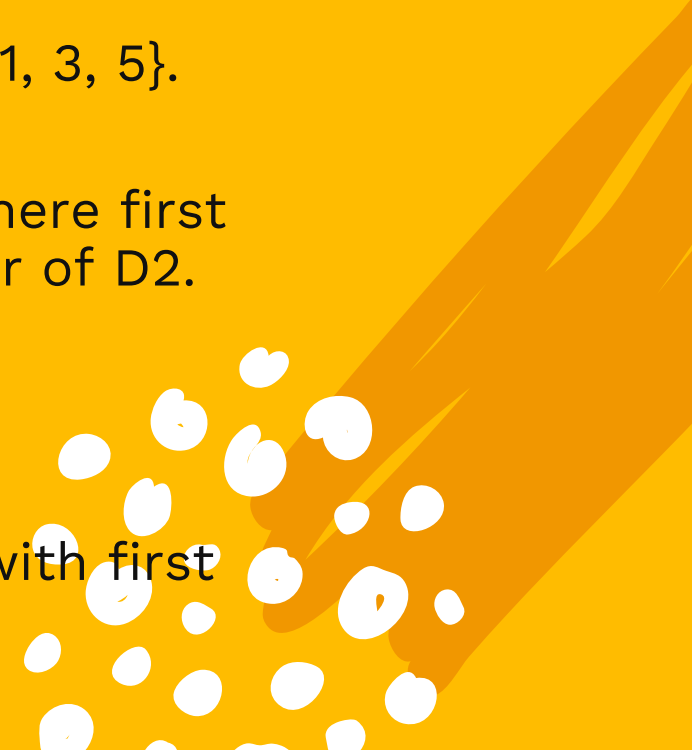


# Alternative Terminology

| Formal terms | Alternative 1 | Alternative 2 |
|--------------|---------------|---------------|
| Relation     | Table         | File          |
| Tuple        | Row           | Record        |
| Attribute    | Column        | Field         |



# Mathematical Relation

- X To understand the true meaning of the term **relation**, we have to review some concepts from mathematics.
  - X Consider two sets, D1 & D2, where  $D1 = \{2, 4\}$  and  $D2 = \{1, 3, 5\}$ .
  - X Cartesian product,  **$D1 \times D2$** , is set of all ordered pairs, where first element is member of D1 and second element is member of D2.
  - X  **$D1 \times D2 = \{(2, 1), (2, 3), (2, 5), (4, 1), (4, 3), (4, 5)\}$**
  - X Alternative way is to find all combinations of elements with first from D1 and second from D2.
- 



X Any subset of Cartesian product is a relation; e.g.

$$\mathbf{R = \{(2, 1), (4, 1)\}}$$

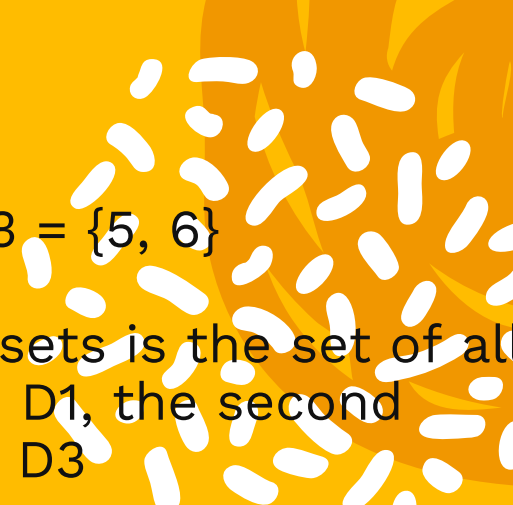
X May specify which pairs are in relation using some condition for selection; e.g.

X second element is 1:

$$\mathbf{R = \{(x, y) \mid x \in D1, y \in D2, \text{ and } y = 1\}}$$

X first element is always twice the second:

$$\mathbf{S = \{(x, y) \mid x \in D1, y \in D2, \text{ and } x = 2y\}}$$

Consider three sets  $D1 = \{1, 3\}$  ,  $D2 = \{2, 4\}$ , and  $D3 = \{5, 6\}$

The Cartesian product  **$D1 \times D2 \times D3$**  of these three sets is the set of all ordered triples such that the first element is from  $D1$ , the second element is from  $D2$ , and the third element is from  $D3$

**$D1 \times D2 \times D3 = \{(1, 2, 5), (1, 2, 6), (1, 4, 5), (1, 4, 6), (3, 2, 5), (3, 2, 6), (3, 4, 5), (3, 4, 6)\}$**

Any subset of these ordered triples is a relation.





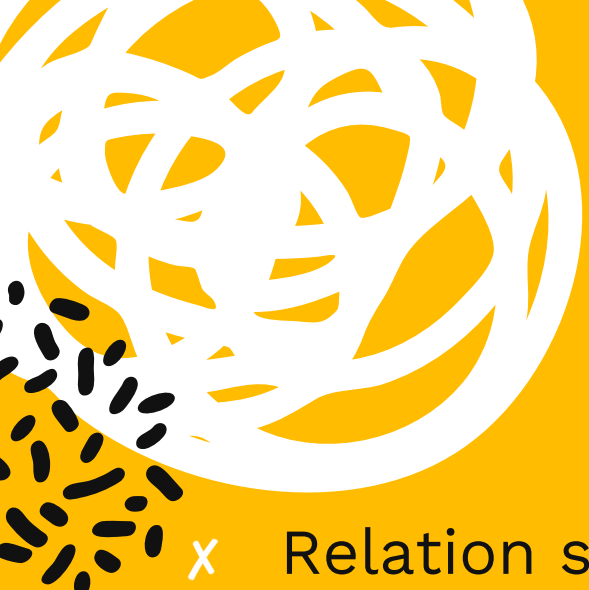
X  $D_1 \times D_2 \times \dots \times D_n = \{(d_1, d_2, \dots, d_n) \mid d_1 \in D_1, d_2 \in D_2, \dots, d_n \in D_n\}$

X and is usually written as:

$$\prod_{i=1}^n D_i$$

X Any set of ***n*-tuples** from this Cartesian product is a relation on the ***n* sets**.





# Database Relation



- x Relation schema: a named relation defined by a set of attribute and domain name pairs
  - x Example of employees' relation schema  
Employees (empid, empname, empemail, empaddress, deptid)
  - x Example of departments' relation schema  
Departments (deptid, deptname, locid)
  - x Example of locations' relation schema  
Locations (locid, locname, locaddress)
- 



# Database Relation

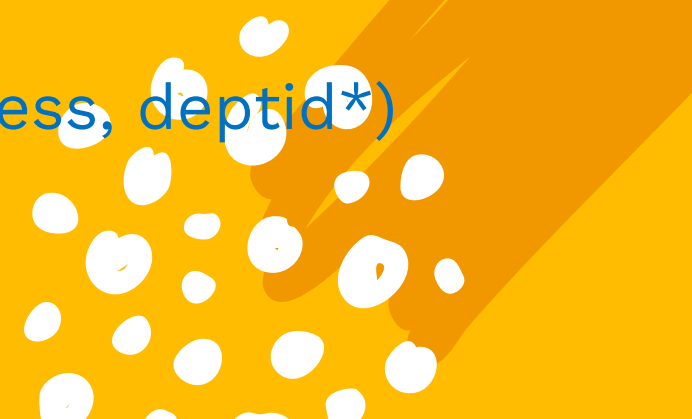


- x Relational database schema: Set of relation schemas, each with a distinct name.
- x Example of Human Resource relational schema which combine of all previous relation schemas

Employees (empid, empname, empemail, empaddress, deptid\*)

Departments (deptid, deptname, locid\*)

Locations (locid, locname, locaddress)





# Relational Keys

- x As stated previously, there are no duplicate tuples within a relation.
- x Therefore, we need to be able to identify one or more attributes (called relational keys) that uniquely identifies each tuple in a relation.

Super  
Key

Candidate  
Key

Primary  
Key

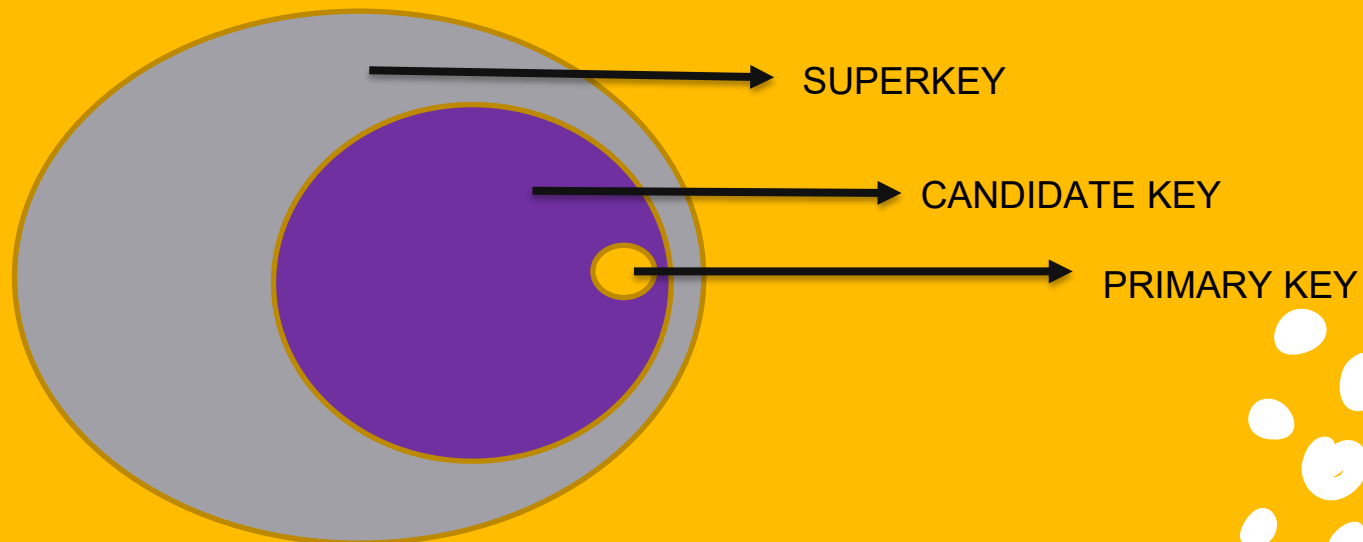
Alternate  
Key

Foreign  
Key



Super key is an attribute, or set of attributes, that uniquely identifies a tuple within a relation.

- x Candidate key is a minimal super key with no redundancy.
  - x Primary key is the candidate key that is selected to identify uniquely within the key relation.
- 



# Example

| vehicleID | carPlateNo | engineID | carName       |
|-----------|------------|----------|---------------|
| 101       | WTY 1234   | 9999876  | Proton Pesona |
| 102       | WXY 4567   | 5644321  | Perodua Alza  |
| 103       | CDA 3389   | 6667889  | Proton Pesona |
| 104       | MCD 1745   | 1277653  | Honda City    |

## Superkey:

- X vehicleID
- X carPlateNo
- X engineID
- X vehicleID, carPlateNo
- X vehicleID, engineID
- X vehicleID, carName
- X carPlateNo, engineID
- X carPlateNo, carName
- X engineID, carName
- X vehicleID, carPlateNo, engineID
- X vehicleID, carPlateNo, carName
- X vehicleID, engineID, carName
- X carPlateNo, engineID, carName
- X vehicleID, carPlateNo, engineID, carName

# Example

| vehicleID | carPlateNo | engineID | carName       |
|-----------|------------|----------|---------------|
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Candidate key:

- X vehicleID
- X carPlateNo
- X engineID
- X vehicleID, carPlateNo
- X vehicleID, engineID
- X carPlateNo, engineID
- X vehicleID, carPlateNo, engineID

Primary Key: vehicleID

# Example

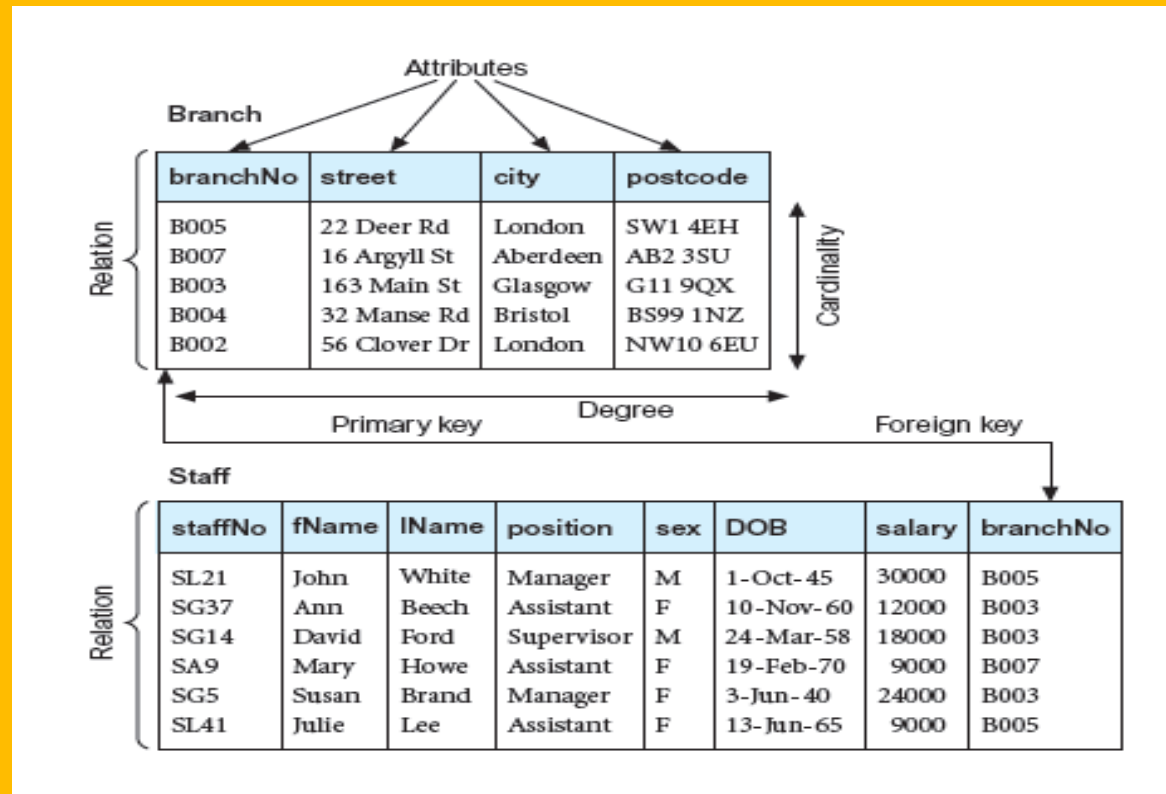
| vehicleID | carPlateNo | engineID | carName       |
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| 103       | CDA 3389   | 6667889  | Proton Pesona |
| 104       | MCD 1745   | 1277653  | Honda City    |

Alternate key:

- X engineID
- X vehicleID, carPlateNo
- X vehicleID, engineID
- X carPlateNo, engineID
- X vehicleID, carPlateNo, engineID

X Alternate Key is a candidate key that was not chosen to be the primary key of the relation.

Foreign Key is an attribute or group of attributes that serves as the primary key of one relation and also appears in another relation (foreign key in this relation).

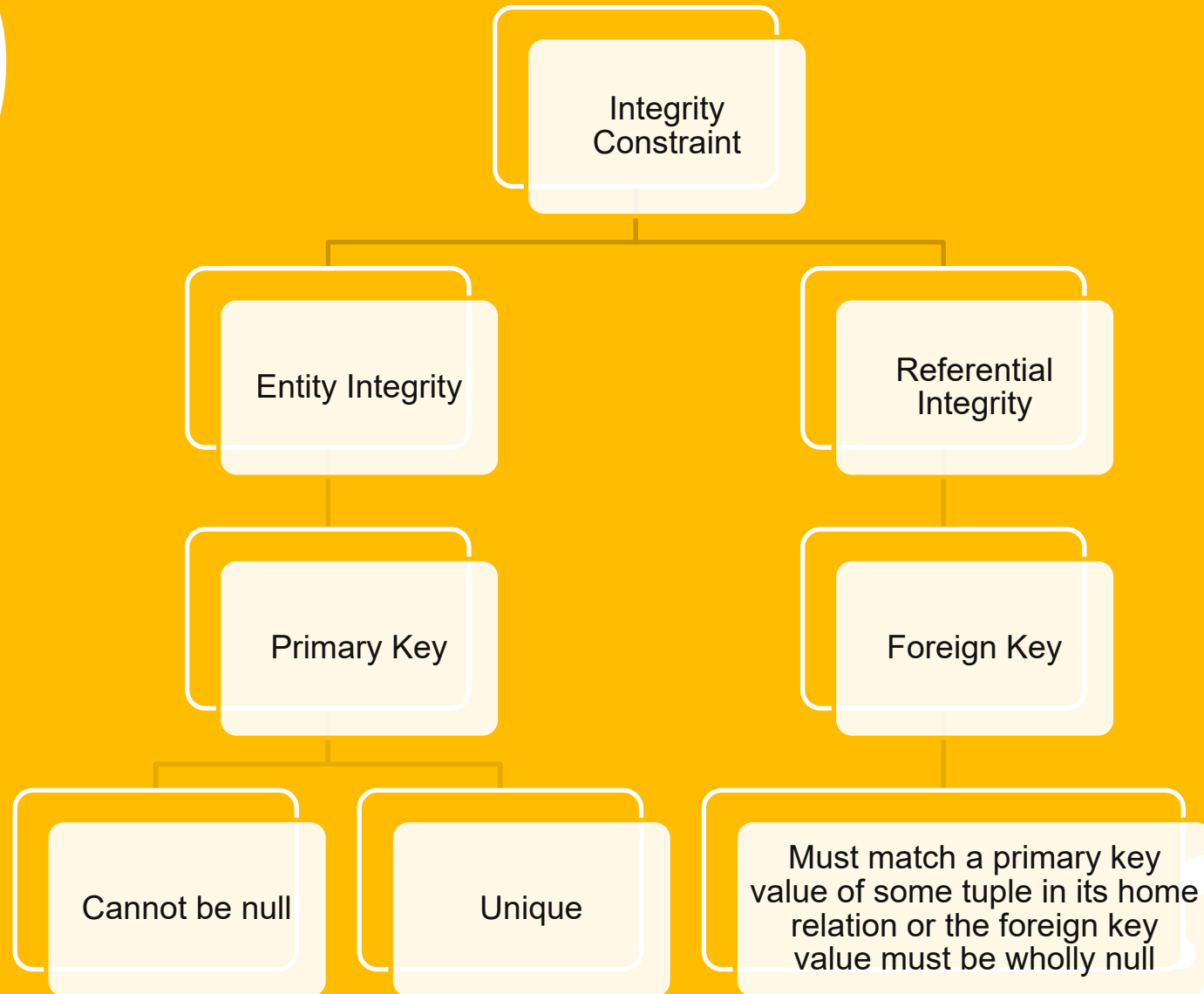




# Integrity Constraints



- X Integrity constraints is to make sure that the data is accurate.
  - X Inaccurate of data could leave the whole database system unreliable and can affect the company from running smoothly.
  - X There are two important integrity rules, which are constraints or restrictions that apply to all instances of the database.
  - X The two principal rules for the relational model are known as entity integrity and referential integrity.
  - X Null value represents a value for an attribute that is currently unknown or is not applicable for this tuple.
- 



# Insert, delete, update for referential integrity

|               | PARENT TABLE                                                    | CHILD TABLE                                                                |
|---------------|-----------------------------------------------------------------|----------------------------------------------------------------------------|
| <b>INSERT</b> | No Problem                                                      | Cannot be done if there is no match primary key in parent's table          |
| <b>DELETE</b> | Cannot be done if there is a match foreign key in child's table | No Problem                                                                 |
| <b>UPDATE</b> | Cannot be done if there is a match foreign key in child's table | Cannot be done if there is no match existing primary key in parent's table |



# Delete & update actions for reference rows



- x Early relational DBMSs did not provide any control mechanisms for referential integrity.
  - x Modern relational DBMSs provide sophisticated control mechanisms for referential integrity:
    - x **Delete rules**
    - x Insert rules
    - x **Update rules**
- 



The **ON DELETE** and **ON UPDATE** action associated with each foreign key in database is one of "**NO ACTION**", "**RESTRICT**", "**SET NULL**", "**SET DEFAULT**" or "**CASCADE**".

- X If an action is not explicitly specified, it defaults to "**NO ACTION**".
  - X **NO ACTION:** Configuring "**NO ACTION**" means just that: when a parent key is modified or deleted from the database, **no special action** is taken.
  - X **RESTRICT:** The "**RESTRICT**" action means that the application is **prohibited** from deleting (for **ON DELETE RESTRICT**) or modifying (for **ON UPDATE RESTRICT**) a parent key when there exists one or more child keys mapped to it.
- 



**SET NULL:** If the configured action is "**SET NULL**", then when a parent key is deleted (for **ON DELETE SET NULL**) or modified (for **ON UPDATE SET NULL**), the child key columns of all rows in the child table that mapped to the parent key are set to contain **SQL NULL** values.

- X **SET DEFAULT:** The "**SET DEFAULT**" actions are similar to "SET NULL", except that each of the child key columns is set to contain the columns **default value** instead of NULL.
  - X **CASCADE:** A "**CASCADE**" action **propagates** the delete or update operation on the parent key to each dependent child key.
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- X For an "**ON DELETE CASCADE**" action, this means that each row in the child table that was associated with the **deleted parent** row is **also deleted**.
  - X For an "**ON UPDATE CASCADE**" action, it means that the values stored in each dependent child key **are modified** to match the **new parent** key values.
  - X **General Constraints: Additional** rules specified by users or database administrators that define or constrain some aspect of the enterprise.
- 



# Views

## **Base Relation**

- Named relation corresponding to an entity in conceptual schema, whose tuples are physically stored in database.

## **View**

- Dynamic result of one or more relational operations operating on base relations to produce another relation.
- 



# Views



- x A virtual relation that does not necessarily actually exist in the database but is produced upon request, at time of request.
  - x Contents of a view are defined as a query on one or more base relations.
  - x Views are dynamic, meaning that changes made to base relations that affect view attributes are immediately reflected in the view
- 

# Purpose of Views

**Provides powerful and flexible security mechanism by hiding parts of database from certain users.**

**Permits users to access data in a customized way, so that same data can be seen by different users in different ways, at same time.**

**Can simplify complex operations on base relations.**



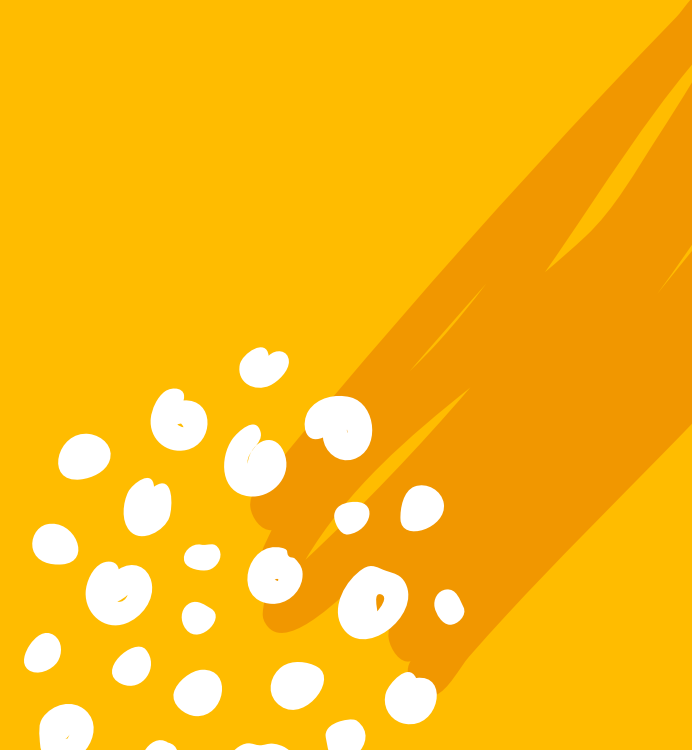
# Updating Views



- x All updates to a base relation should be immediately reflected in all views that reference that base relation.
  - x If view is updated, underlying base relation should reflect change.
  - x Modification's restrictions:
    - x Updates are allowed if query involves a single base relation and contains a candidate key of base relation.
    - x Updates are not allowed involving multiple base relations.
    - x Updates are not allowed involving aggregation or grouping operations.
- 



# Classes of Views

- x theoretically not updateable;
  - x theoretically updateable;
  - x partially updateable.
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# References

- x Thomas Connolly and Carolyn Begg, Database Systems: A Practical Approach to Design, Implementation, and Management, 6th Edition, Pearson, 2015, ISBN: 978-01329432
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